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RESEARCH ARTICLE

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Projected shifts in dry–wet extremes and possible associated drivers over the Hindu Kush Himalaya

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ABSTRACT

The Hindu Kush Himalaya (HKH) region is characterized by its mountainous terrain and diverse regional climate patterns. This study investigates how future hydroclimate extremes across four regions of the HKH may be influenced by various Shared Socioeconomic Pathways (SSPs) scenarios, using a CMIP6-based Multi-Model Ensemble (MME). Our results show that the frequency of dry days is projected to increase, with a notable shift in more extreme spells over the extended parts of the northwest Himalaya and Karakoram regions (Zone 1) under high emissions. The eastern HKH (Zone 4) is projected to experience substantially increased precipitation with standardized precipitation index (SPI) values of 1.5–3, indicating moderately wet to extremely wet conditions that are likely to intensify flood risk. In consonance with the drier near-future climate over HKH (Zone 3), we find that the mean local Hadley cell shows a widening signature (expanding to 33.5° latitude). However, the Hadley cell shows an equatorward shift and narrowing signatures during mid (25.6° latitude) and far future (27.7° latitude) under low-to-high emissions scenarios, thus exemplifying the wet conditions over HKH (Zone 3). This work highlights the pressing need to understand the spatial heterogeneity of future global warming impacts over HKH.

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Hindu Kush Himalaya (HKH); hydroclimate extremes; CMIP6; standardized precipitation index (SPI); ITCZ

1. Introduction

The Hindu Kush Himalaya (HKH) region, often referred to as the water tower of Asia, supports nearly two billion people, approximately one quarter of the world's population, and has been identified by the Intergovernmental Panel on Climate Change (IPCC) as a hotspot of climate change in recent decades (Wester et al. 2019). The HKH acts as a water source for highly elevated South Asian river basins, such as the Ganga, Indus, Brahmaputra River Basins, Yellow, and Yangtze, and hence plays a vital role in ensuring that water resources reach more than 1.5 billion people (e.g., Wester et al. 2019; Immerzeel et al. 2020; Kad and Ha 2023; Chaubey et al. 2024). The HKH region experiences extreme precipitation linked to an intensified monsoon circulation and a subtropical westerly jet (e.g., Sandu et al. 2019; Nischal et al. 2024). The Himalayas are the fastest retreating glaciers in the world, with significant ice loss in the region's glaciers over the past few decades (Hock et al. 2019; Shukla and Sen 2021; Naegeli et al. 2022). Therefore, it is important to understand the pattern of future climate extremes to examine how such changes may influence hydrology and hydroclimate extremes, such as drought and intense precipitation, over the HKH. Given these vulnerabilities, robust projections of future climate extremes are essential for informing regional adaptation strategies.

Previous researchers used different Representative Concentration Pathway (RCP) scenarios of Coupled Model Intercomparison Project phase 5 (CMIP5) to examine the observed and future changes in weather extremes such as floods and droughts over the HKH (e.g., Sanjay et al. 2017; Krishnan et al. 2020). In the

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present study, we employ the IPCC Shared Socioeconomic Pathways (SSPs) from Coupled Model Intercomparison Project Phase 6 (CMIP6; Eyring et al. 2016) to analyse future changes in hydrological extremes, incorporating diverse socioeconomic dimensions, including sustainability, inequality, and fossil fuel development (e.g., Riahi et al. 2017). This enhanced framework facilitates an integrated analysis of regional vulnerabilities, as well as adaptation and mitigation strategies. Understanding these future changes requires not only improved climate scenarios but also robust indices to quantify hydroclimate extremes.

Extreme weather events, such as heavy rainfall and droughts, pose significant risks to human-made ecosystems and natural systems (Giorgi et al. 2018). Moreover, South Asia experienced approximately 164 flood events in 2023, affecting approximately 10.2 million people in India alone (CRED 2023). Many previous researchers have used consecutive dry and wet days to examine the intensity and frequency of extreme precipitation (e.g., Breinl et al. 2020; Du et al. 2022; Chaubey and Mall 2023). The Standardised Precipitation Index (SPI) is also suggested as an alternative approach to examine the long-term dry-to-wet conditions over the region (McKee et al. 1993). Recent studies have also used Generalized extreme value (GEV) distribution based on annual maxima theory to examine the projections of precipitation extremes and the return period (Mannshardt-Shamseldin et al. 2010). Vertical velocity (VV) plays an important role in precipitation analysis because it determines the rate at which air rises, which is essential for lifting moisture to altitudes where it can condense and form clouds, ultimately leading to precipitation (Gimeno-Sotelo et al. 2024). Therefore, the combined analysis of VV and precipitation provides a robust framework for assessing and understanding changes in extreme precipitation using climate models. However, interpreting changes in precipitation extremes requires understanding their underlying physical drivers. Increasingly, it has also been noted that Soil moisture (SM) significantly influences water vapour availability as well as atmospheric stability and circulation by modifying the surface energy balance, affecting precipitation patterns in distant locations (e.g., Attada et al. 2022; Hua et al. 2022; Qiao et al. 2022). The Intertropical Convergence Zone (ITCZ) also plays a crucial role in determining regional precipitation patterns. Recently, Kad and Ha (2023) have examined ITCZ changes during the observed period (1979–2021) and their impacts on HKH precipitation extremes using reanalysis data. However, the role of ITCZ on precipitation extremes has not been widely studied, particularly in South Asia (Byrne and Schneider 2016; Hari et al. 2020) and the HKH regions.

Although previous studies have examined precipitation changes over South Asia, a detailed investigation of dry-wet extremes and their associated drivers, specifically over the HKH region using the low to high emission SSPs scenarios, remains limited. Building on these identified research gaps, our study focuses on two key scientific objectives: (1) investigating the projected changes in dry-wet extremes over the HKH during the near, mid and far future periods under different emission scenarios using CMIP6 climate model outputs, and (2) identifying the potential factors driving these future changes in hydroclimate extremes.

2. Data and methods

2.1. Study area

The Hindu Kush Himalayan (HKH) region extends from approximately 62°E–103°E longitude and 27°N–39°N latitude, encompassing one of the most complex and high-elevation terrains on Earth, with elevations ranging from approximately 500 m to over 8600 m (Wester et al. 2019). The region supports nearly two billion people and serves as the water tower of Asia through its major river systems, including the Indus, Ganga, Brahmaputra, Yellow, and Yangtze. The HKH region exhibits a strong west-to-east precipitation gradient: the western HKH region receives moisture primarily from extratropical westerly disturbances during winter and spring, while the eastern HKH region is dominated by the South Asian summer monsoon (June–September) (Krishnan et al. 2019). In this study, the HKH domain was divided into four sub-regional zones following Sanjay et al. (2017), based on their distinct hydroclimate characteristics: Zone 1: Extended Part of the Karakoram (62°–71°E, 28°–36°N), Zone 2: the Northwest Himalayan region representing HKH1 (71°–79°E, 32°–39°N), Zone 3: the Central Himalayan region indicates HKH2 (76°–93°E, 27°–32°N), and Zone 4: the Southeast Himalayan and Tibetan Plateau

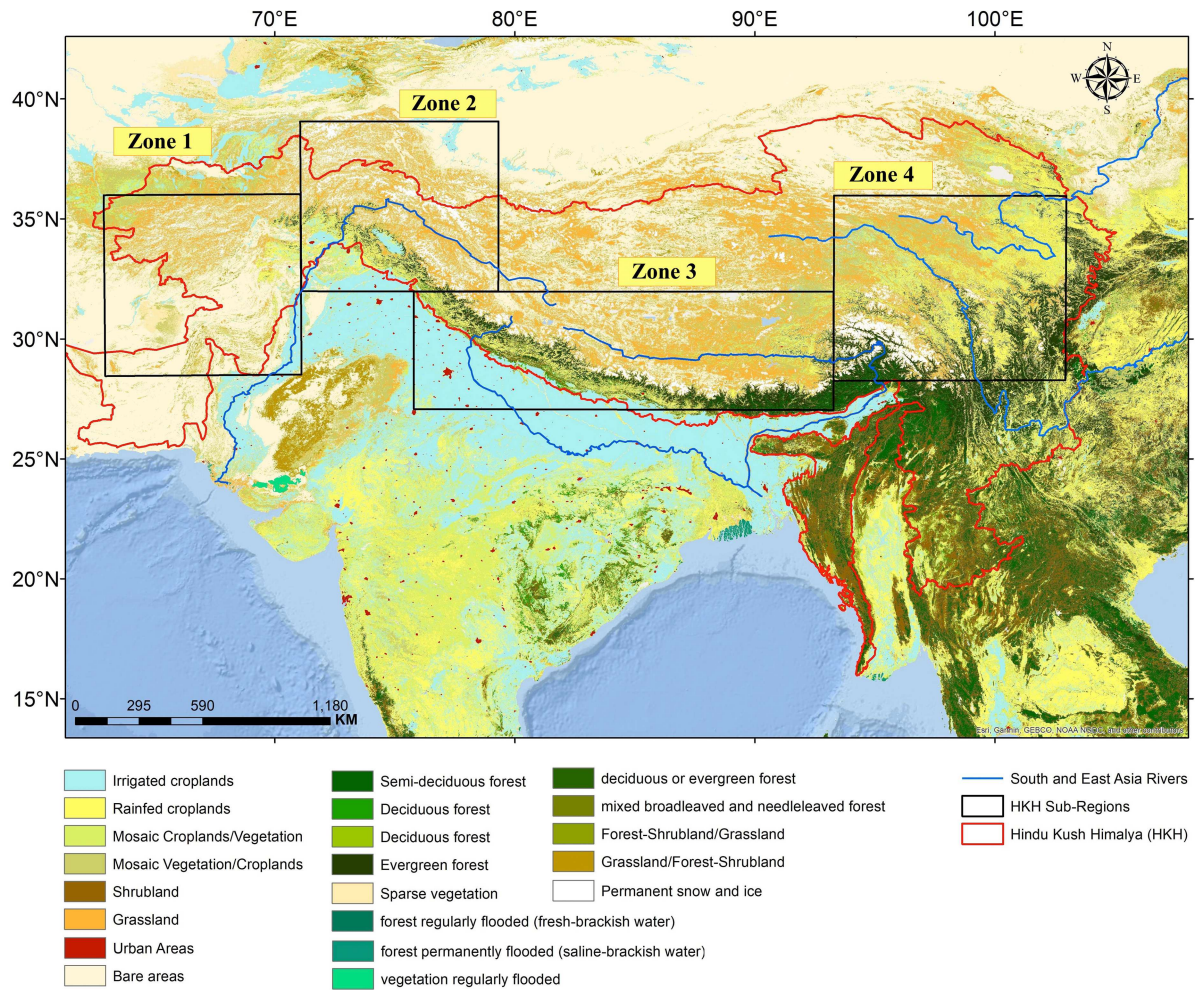


Figure 1. Study area reflects Hindu Kush Himalaya (HKH) region cover by four zones namely, Zone 1: extended part of the Karakorum (South-western part of the HKH; 62°–71°E, 28°–36°N), Zone 2: northwest Himalaya and Karakoram (71°–79°E, 32°–39°N), Zone 3: central Himalaya (76°–93°E, 27°–32°N), and Zone 4: southeast Himalaya and Tibetan Plateau (93°–103°E, 28°–36°N). The respective land use and land cover data are shown using ESRI and NOAA Satellite imagery.

represents HKH3 (93°–103°E, 28°–36°N) (Figure 1). The selection of these zones is based on the previous research (Sanjay et al. 2017) to examine the zonal-specific climate responses over HKH. The spatial distribution of these zones is shown in Figure 1.

2.2. Observation and CMIP6 data

We used the daily gauge-based gridded observational datasets APHRODITE's (Asian Precipitation – Highly-Resolved Observational Data Integration Towards Evaluation) at high resolution ($0.25^\circ \times 0.25^\circ$) for the period of 1951–2014 to examine the precipitation characteristics over four zones of the HKH. Projected precipitation characteristics were analyzed using the historical (1951–2014) and Shared Socioeconomic Pathways (SSPs) experiments from sixteen coupled climate models contributing to CMIP6 (Eyring et al. 2016; O'Neill et al. 2020), as organized by the World Climate Research Programme (WCRP). We considered the lowest-to-highest forcing scenarios for SSPs: i.e., SSP1-2.6 (low-emission scenario), SSP2-4.5 (medium-emission scenario), SSP3-7.0 (high-emission scenario), and SSP5-8.5 (highest-emission scenario). All variables of the CMIP6 datasets are used with the variant label r1i1p1f1. All the datasets are interpolated to a common spatial resolution of one degree. To reduce the uncertainty in the climate model datasets, bias correction was performed using the quantile mapping technique, which matches the cumulative distribution functions of model outputs with APHRODITE

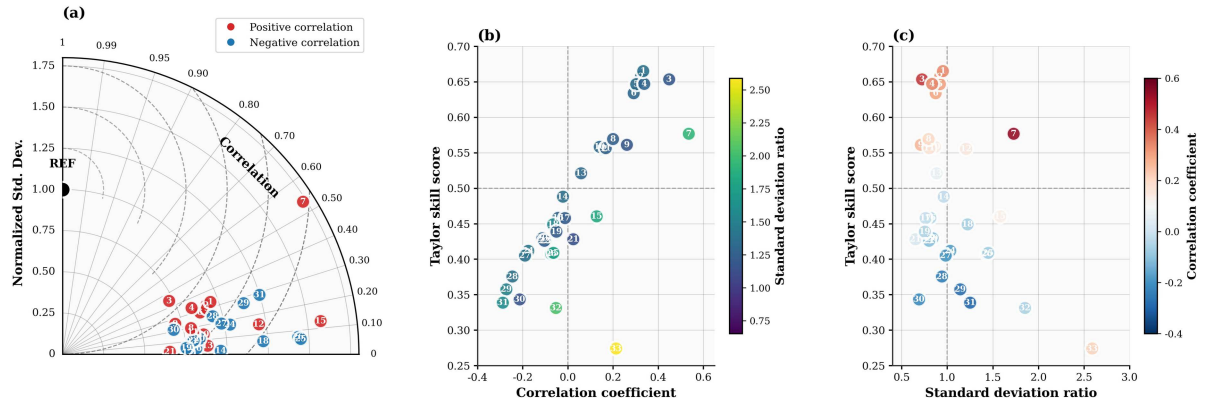


Figure 2. Statistical comparison of thirty-three CMIP6 models with the APHRODITE reference dataset for the HKH region (1951–2014). The figure shows the (a) Taylor diagram summarizing model correlation with observed datasets, (b) relationship between the Taylor Skill Score (TSS) and correlation, with points colored by their standard deviation ratio, indicates model skill varies with correlation and (c) the relationship between the TSS and the standard deviation ratio, with points colored by their correlation coefficient, indicates model skill varies with variance representation. The model-ranked list based on their TSS is shown in Table 1.

observational data for the historical period (1951–2014), following the methodology of Xu et al. (2021), and model skill was evaluated using the Taylor Skill Score (TSS; Xu et al. 2021; Li and Zhong 2025). The results are presented using the Multi-Model Ensemble (MME) method, which is based on bias-corrected positive correlation model datasets (Figure 2a–c; Supplementary Table S1 & Figures S1–S3). We examine future changes in precipitation extremes, SM, and VV over HKH for the near-future (2021–2040), mid-future (2041–2060), and far-future (2081–2100) at all four regions, according to the time frame defined by the IPCC AR6 report and previous research (IPCC et al. 2021; Jin et al. 2025).

2.3. Estimation of soil moisture (SM) and precipitation extremes

To illustrate SM over the HKH, soil moisture at the 10 cm surface layer was used to characterize near-surface water availability. This layer responds most rapidly to precipitation anomalies and is the standard CMIP6 output used for agricultural drought assessment in high mountain and Central Asian regions (Qiao et al. 2022; Hua et al. 2022). The Standardized Soil Moisture Index (SSMI) was computed as a normalised measure of the soil moisture anomaly, indicating how individual-year soil moisture deviates from the long-term mean.

$$SSMI = [(SM_{\text{Individual year}} - SM_{\text{long term Avg.}})/SM_{\text{long term Standard deviation}}] \quad (1)$$

The Soil Water Index (SWI) measures soil moisture at different depths, influenced by precipitation through the process of infiltration. A drought or water scarcity conditions are typically identified when the soil moisture percentile falls below 20%. SWI was calculated from the equation given below:

$$SWI_p = [(R - 0.5)/N \times 100] \quad (2)$$

where SWI_p = Percentile for the SWI value, R = period of interest (monthly), N = total number of periods.

Projected percentage changes in soil moisture were evaluated following the IPCC 2021 methodology for the near-, middle-, and far-future periods relative to the baseline period of 1995–2014, using the equation below:

$$\%SM = [(SM_n - SM_m)/SM_m] * 100 \quad (3)$$

here, SM indicates that the soil moisture, n is the particular year soil moisture values, and m is the long-term average of soil moisture.

To examine changes in extreme events over the HKH, we applied the Expert Team on Climate Change Detection and Indices (ETCCDI) across low- to high-emission scenarios (Table 1). These indices,

Table 1. Definitions of the expert team on climate change detection and indices (ETCCDI).

ETCCDI indices name	Indices	Unit	Definition
Consecutive Dry Days	CDD	Days	Maximum length of dry spell, maximum number of consecutive days with $RR < 1\text{mm}$
Consecutive Wet Days	CWD	Days	Maximum length of wet spell, maximum number of consecutive days with $RR \geq 1\text{mm}$

Here, PR = precipitation.

Table 2. SPI classification was applied in this study.

S. No.	SPI (McKee et al. 1993)	Description
1.	≥ 2.00	Extremely wet
2.	1.50 to 1.99	Very wet
3.	1.00 to 1.49	Moderately wet
4.	0.50 to 0.99	Slightly wet
5.	-0.49 to 0.49	Near Normal
6.	-0.99 to -0.50	Slightly dry
7.	-1.49 to -1.00	Moderately dry
8.	-1.99 to -1.50	Very dry
9.	≤ -2.00	Extremely dry

including Consecutive Dry Days (CDD) and Consecutive Wet Days (CWD), were used to estimate the intensity and frequency of extreme events in warmer futures, focusing on dry and wet conditions. For drought assessment over the HKH, the Standardized Precipitation Index (SPI) was computed at the 12-month time scale using the NCL (NCAR Command Language) function `dim_spi_n`, which implements the standard McKee et al. (1993) methodology. At each grid point, the monthly precipitation time series over the historical period (1951–2014) is fitted to a two-parameter gamma distribution with shape parameter α and scale parameter β , estimated via maximum likelihood. The cumulative gamma probability $G(P_k)$ is then computed and transformed to the standard normal distribution using the inverse transform. The SPI ranking is given in Table 2, and the fitted gamma distribution to monthly precipitation (P_k) is defined below in Equation (4):

$$f(x) = \left(\frac{1}{\beta^\alpha \Gamma(\alpha)} \right) x^{\alpha-1} e^{-x/\beta} \quad (4a)$$

Here, x = precipitation amount, α = shape parameter, β = scale parameter, $\Gamma(\alpha)$ = gamma function:

$$G(p_k) = \int_0^{p_k} f(x) dx \quad (4b)$$

$$SPI = \phi^{-1}[G(p_k)] \quad (4c)$$

where ϕ^{-1} is the inverse normal cumulative distribution function.

The return levels of projected daily precipitation extremes were estimated using a Generalised Extreme Value (GEV) distribution fitted to annual maximum precipitation at each grid point. The GEV is characterized by three parameters: location (μ), scale (σ), and shape (ξ), which are all estimated via maximum likelihood estimation (MLE). The 10-year return level (corresponding to a 10% annual exceedance probability, i.e. at least one occurrence per decade) was used to assess the intensity of future extremes. The statistical significance of the projected changes in return levels was evaluated using the likelihood-ratio test ($p < 0.05$). Full methodological details are provided in Chaubey and Mall (2023).

We extract the VV at 500 hPa ($\omega = dp/dt$, where p refers to daily atmospheric pressure) ($\text{Pa}\cdot\text{s}^{-1}$) from CMIP6 for historical and all SSP scenarios. In this study, for any stable weather, we consider positive values of " ω ", which indicate downward motion; accordingly, " $-\omega$ " denotes updraft motion. Here, we consider the VV at 500 hPa, as it represents the mid-troposphere above the HKH terrain, effectively capturing large-scale vertical motion associated with synoptic systems, including western disturbances and extratropical cyclones, that drive precipitation extremes in this high-elevation region (Nischal et al. 2024).

2.4. Local Hadley circulation, ITCZ location, and its width

The local meridional mass stream function was used to examine the mean meridional circulation over the HKH domain (Peixoto and Oort 1984; Kad and Ha 2023). The stream function was computed by zonally averaging the meridional velocity over the HKH longitudinal sector (60°E–100°E) and integrating from the surface to the top of the atmosphere, across the latitudinal range 0°N–40°N, thereby capturing the full extent of the local Hadley cell relevant to HKH precipitation dynamics. Here, the local meridional mass stream function is defined as below:

$$\Psi = \left(\frac{2 \pi a \cos \phi}{g} \right) * \int_P^{P_s} (\bar{V}) dp \quad (5)$$

here, Ψ is the zonal mean meridional velocity, a indicates the radius of the Earth, g specifies the acceleration due to gravity, and P_s denotes the surface pressure.

The ITCZ location is defined as the latitude closest to the equator where the zonal mean stream function, vertically averaged between 700 and 300 hPa, is found to be zero (Byrne and Schneider 2016). The formulation is given below:

$$\phi_{ITCZ} = \phi_{\Psi=0} \quad (6)$$

The ITCZ width is characterized by the latitudinal distance between the maximum and minimum points, which is determined using the zonal mean stream function that is vertically averaged between 700 and 300 hPa. The ITCZ width is evaluated using the following equation:

$$\phi_{width} = (\phi_{max} - \phi_{min}) \quad (7)$$

3. Results

3.1. Spatial distribution of precipitation and soil moisture

The complex topography of the Himalayas exerts a profound influence on the spatial distribution of precipitation across the HKH region, giving rise to marked west-to-east heterogeneity. In this section, we examine the spatial patterns of both precipitation and soil moisture over the whole HKH domain, with an additional focus on each of the four sub-regional zones.

3.1.1. The whole region (HKH)

Figure 3 illustrates the projected daily mean precipitation for the HKH region between 2021 and 2100, under a suite of SSP scenarios spanning low to high emissions. Historically, the areal annual mean precipitation was approximately 2.59 mm/day, based on data from the 1951–2014 reference period. Under future projections, the ensemble mean rises to 2.68 mm/day for the low emission (SSP1-2.6) scenario and 3.05 mm/day under the high emission (SSP5-8.5) pathway, reflecting a consistent intensification of precipitation as greenhouse gas emissions increase. The moisture content, historically confined between 0 and 40 kg m⁻², is projected to increase substantially under all future scenarios, with maxima approaching 70 kg m⁻² (Figure S4). The western HKH region is projected to experience a reduction in annual precipitation, in stark contrast to the eastern sector, which exhibits a robust increasing trend (Figure 4a). Moreover, in future projections, the annual mean precipitation will increase more under SSP2-4.5 and SSP5-8.5 (Figure 4b). As shown in Figure 5, the spatial distribution of the percentage changes in precipitation and soil moisture relative to the 1995–2014 baseline were derived from IPCC AR6. In terms of percentage change, under SSP2-4.5 and SSP5-8.5, much of the HKH region experiences weak or inconsistent signals of change in the near future, with marginal increases primarily emerging in the far-future period. Notably, projections for the near future (–15% to +10%) and mid-century (–10% to +15%) exhibit substantial intermodel spread, underscoring persistent uncertainties in regional hydrological responses to anthropogenic forcing.

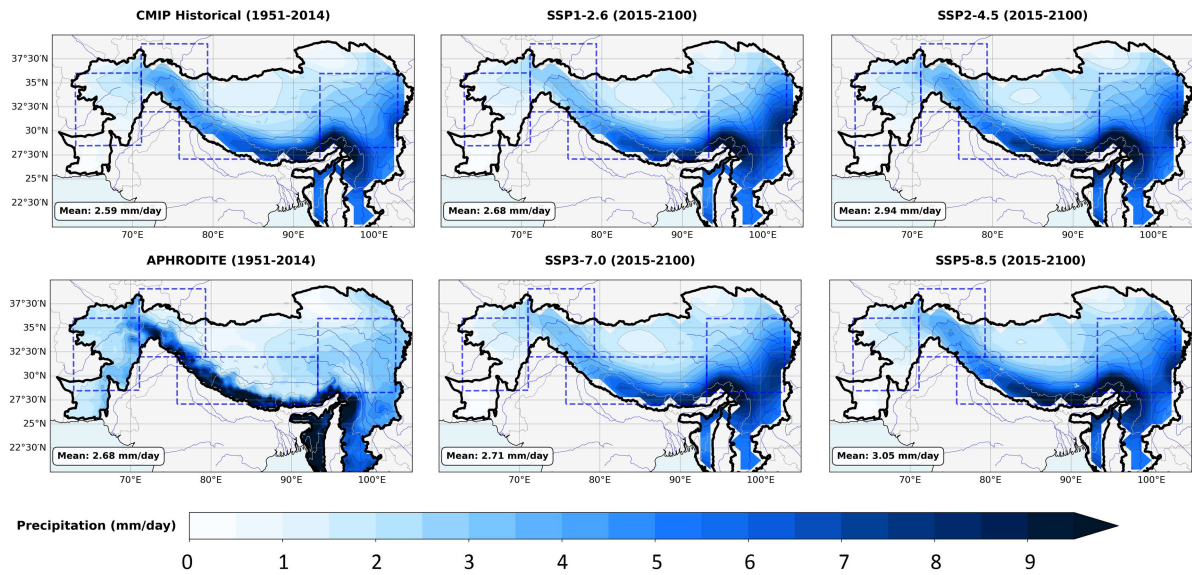


Figure 3. Climatological daily mean precipitation (in mm/day) for CMIP6-Historical and Aphrodite for the period 1951–2014 and projection for the period 2015–2014.

3.1.2. Extended part of the Karakoram (zone 1)

Zone 1, representing the extended Karakoram sector of the HKH, exhibits a persistent deficit in daily mean precipitation, historically ranging from 0 to 2 mm/day, in contrast to estimates from the APHRODITE dataset, which indicate higher values of up to 5 mm/day (Figure 3). A similar deficiency ($0\text{--}20\text{ kg m}^{-2}$) is evident in the historical soil moisture distribution, particularly under low-emission scenarios (SSP1-2.6 and SSP2-4.5) (Figure S4). However, projections suggest a notable increase in moisture content, reaching up to 30 kg m^{-2} , under higher emission scenarios. The precipitation projected trends across the region reveal a weak negative slope, varying from -4.6×10^{-2} to 2×10^{-2} mm/day/decades, indicative of declining precipitation (Figure 4). Furthermore, a substantial reduction in precipitation ranging from 0% to -30% is projected across all future periods under both the SSP1-2.6 and SSP3-7.0 scenarios (Figure 5), underscoring the region's vulnerability to drying in the future.

3.1.3. Northwest Himalayan (zone 2)

Zone 2, covering the northwestern Himalaya and adjacent Karakoram sector, is characterized by steep topography and complex orographic influences. Historical records indicate daily mean precipitation ranging from 2 to 6 mm/day, while the APHRODITE dataset suggests higher intensities, reaching up to 9 mm/day (Figure 2). In contrast, soil moisture across the region exhibited minimal spatial variation, with historical values of approximately 20 kg m^{-2} and future projections showing a modest increase to $20\text{--}30\text{ kg m}^{-2}$ across scenarios (Figure S4). Precipitation trends are found to be increasing under increasing emission scenarios, indicating a potential intensification of hydrological input (Figure 4). The projected changes in precipitation under SSP2-4.5 and SSP5-8.5 suggest a moderate rise in precipitation, with changes ranging from 0% to 10% across future time horizons (Figure 5).

3.1.4. Central Himalayan (zone 3)

Zone 3, representing the central Himalayan sector of the HKH, displays pronounced variability in precipitation patterns due to its complex terrain and dynamic monsoonal influence. Projections of daily mean precipitation reveal substantial spatial heterogeneity, ranging from 3 to 6 mm/day across the region (Figure 3). In contrast, soil moisture exhibited a relatively uniform distribution, with values spanning $0\text{--}30\text{ kg m}^{-2}$ under both historical and high-emission scenarios (Figure S4). The projected trends of precipitation indicate a robust increase, particularly over the high-elevation zones, where intensification rates exceed 2×10^{-2} mm/day/decade (Figure 4). This upward trend is further supported by statistically significant increases in ($\geq 13 \times 10^{-2}$ mm/day/decade) precipitation under the high-emission SSP5-8.5

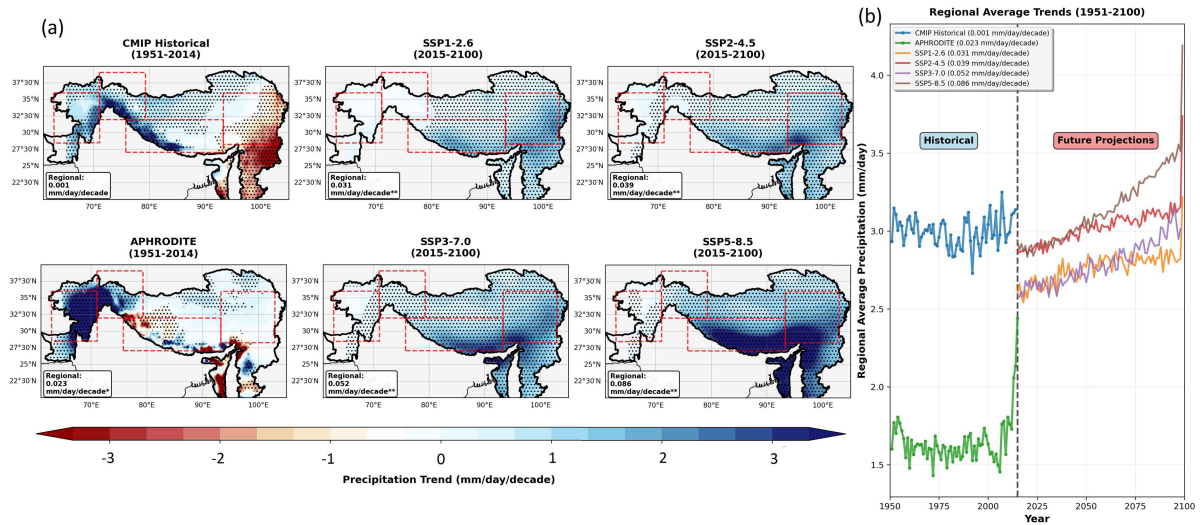


Figure 4. The spatio-temporal trend of (a) daily mean precipitation per decade for low to highest emission scenarios, (b) annual mean precipitation for the HKH region.

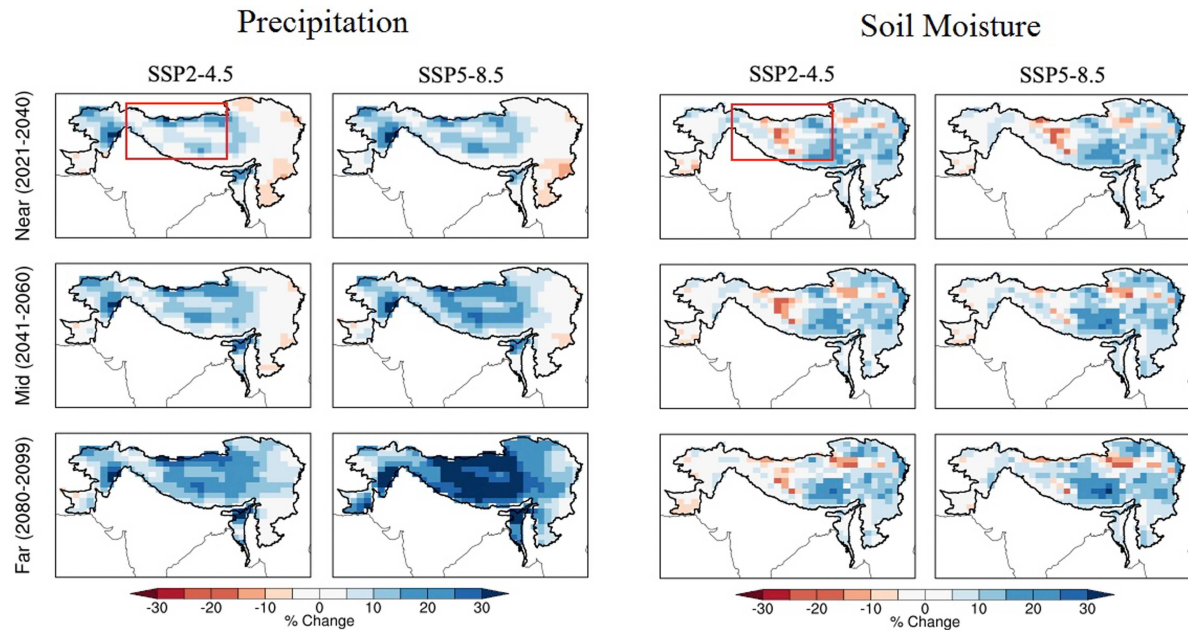


Figure 5. Spatio-temporal percentage changes in the annual mean precipitation and soil for near- (2021–2040), mid- (2041–2060), and far-future (2081–2100) relative to 1995–2014. The red box indicates the decline of precipitation and soil moisture in the near future.

scenario ($p \leq 0.05$). Under ongoing warming, Zone 3 is projected to experience a marked amplification in precipitation, with increases ranging from 5% to over 40% in the far future under the SSP2-4.5 and SSP5-8.5 scenarios (Figure 5), highlighting its critical role in shaping downstream water availability and flood risk under the future climate. While in the near future, this zone shows the normal and declining soil moisture changes -20% to -10% , which indicates that the dry spells over the region.

3.1.5. Southeast Himalayan and Tibetan Plateau (zone 4)

Zone 4, encompassing the southeast Himalaya and the adjoining Tibetan Plateau, registers the highest annual mean precipitation across the HKH, with historical values exceeding 7 mm/day, while this zone is

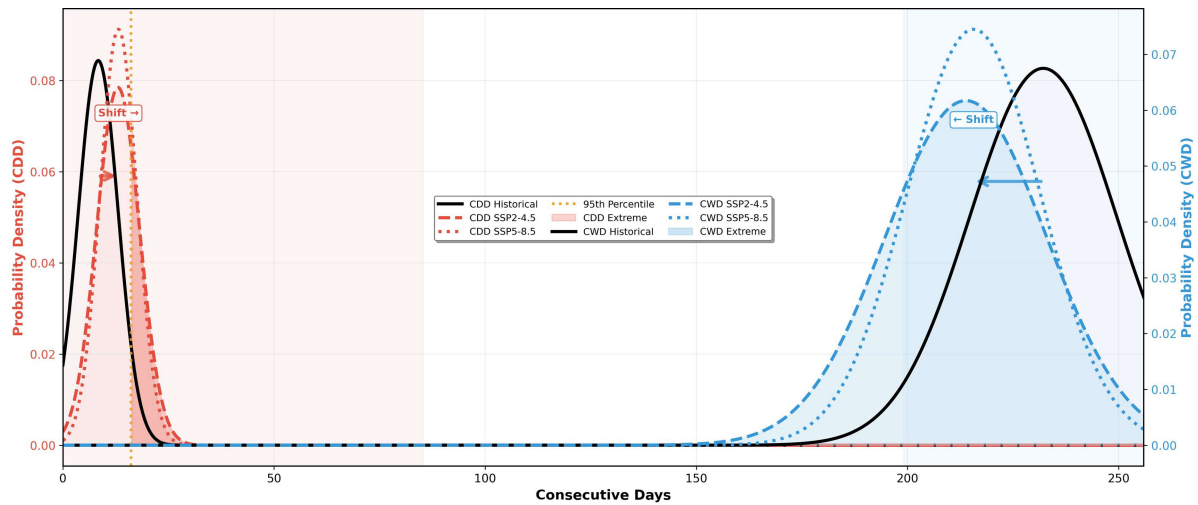


Figure 6. Projected shifts in the dry and wet days using probability distributions for Central Himalaya (Zone 3) for the near future (2021–2040). The shift in the peaks indicates a future trend towards longer dry spells and shorter wet spells. Shaded areas represent extreme conditions, with the 95th percentile of historical CDD marked by the dotted line.

projected to be ≥ 9 mm/day (Figure 2). This region also exhibits the greatest soil moisture content, which historically ranges from 20 to 50 kg m⁻², and in future scenarios reaching 50–70 kg m⁻² under the high-emission SSP5-8.5 scenario (Figure S4). The rise in moisture content is consistent across the emission spectrum, indicating a strong coupling between warming and atmospheric moisture availability. The precipitation trends over this zone reveal a statistically significant ($p \leq 0.05$) and rapid increase under SSP5-8.5, with rates varying spatially from $\geq 4 \times 10^{-2}$ mm/day/decade (Figure 4). However, the projected precipitation changes exhibit a non-linear temporal evolution: moderate declines (–10% to –15%) are observed during the near- and mid-future periods, followed by a notable intensification (5% to 30%) in the far future under SSP5-8.5 (Figure 5).

3.2. Projected changes in extremes

In this section, we compute the projected shift in the CWD and CDD over the HKH region. We found that Zone 3 shows the dry days increasing shift in the near future, in comparison with the historical period of datasets (Figure 6). The spatial projected changes in extremes over the HKH in terms of the frequency and intensity of consecutive dry and wet days are shown in Supplementary Figures S7 and S8. The Zone 3, situated mainly in the Himalayan range, shows a reduced number of projected wet days in comparison to historical, and it is found to vary from 150 to 400 days. The probability distributions of dry and wet days for the Zone 3 in the near future reveal pronounced shifts compared to historical conditions. Future scenarios exhibit a clear shift toward longer dry spells, with the peak probability density of CDD moving to higher values and surpassing the 95th percentile of historical extremes, indicating an increased risk of prolonged dry events. Conversely, the distribution of wet spells contracts, with the peak shifting toward fewer consecutive wet days, suggesting a decrease in the duration of extended wet periods (Figure 6).

Figure S8 shows the projected change in the intensity of the wet and dry events (Figure 7a). Zone 4 is projected to experience a substantial increase in consecutive wet days, rising from 400 to 600 days across low-to-high emission scenarios (Figure S8), indicating a marked intensification of prolonged wet spells and elevated flood risk in the far future. The Zone 1 and Zone 2 exhibit the highest spatial distribution of extreme precipitation (200–400 mm/day) under the low-emission scenario (SSP1-2.6) and moderately high-emission scenario (SSP3-7.0), with a 10-year return period (Figure 7a). In contrast, Zones 3 and 4 show enhanced 10-year return levels of 100–250 mm/day under the medium (SSP2-4.5) and high (SSP5-8.5) emission scenarios. These shifts in spatial maxima indicate a redistribution of extreme precipitation events toward the eastern HKH under stronger warming. Moreover, the outliers in Figure 7b represent projected extreme precipitation events over the HKH region.

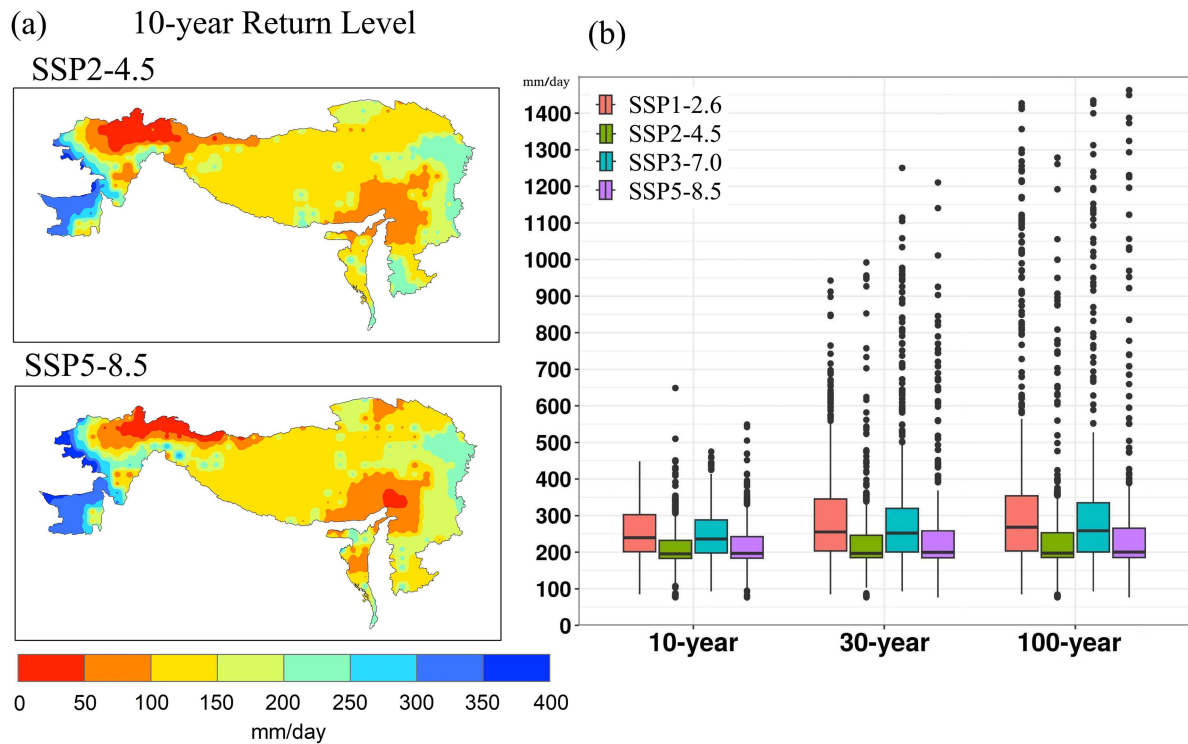


Figure 7. Estimation of the return level of precipitation extremes over HKH, (a) spatially at 10-year return levels (10%, probability storms), (b) boxplot for 10-, 30- and 100-year return levels using 2021–2100 projected SSP scenarios.

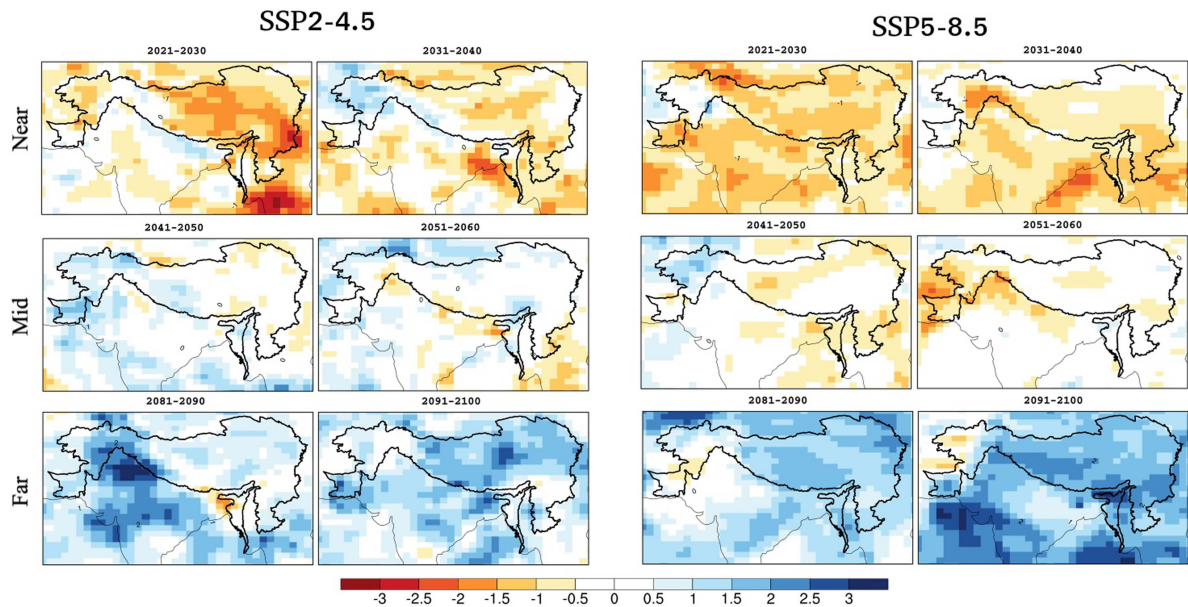


Figure 8. Decadal changes in the average intensity of projected wet-to-dry conditions at a 12-month time span over HKH in medium and high emission SSPs scenarios from 2021 to 2100.

Figure 8 shows the decadal changes in the average intensity of the estimated SPI at a 12-month time scale. The Zones 3 and 4 regions exhibited slightly dry to very dry ($-2.5 \leq \text{SPI} \leq -0.5$) intensities in the near future during the decades 2021–2030 and 2031–2040 in all SSP scenarios (Figure 8). However, the mid-future shows mixed signatures across the scenarios. Under the medium-emission scenario SSP2-4.5, the

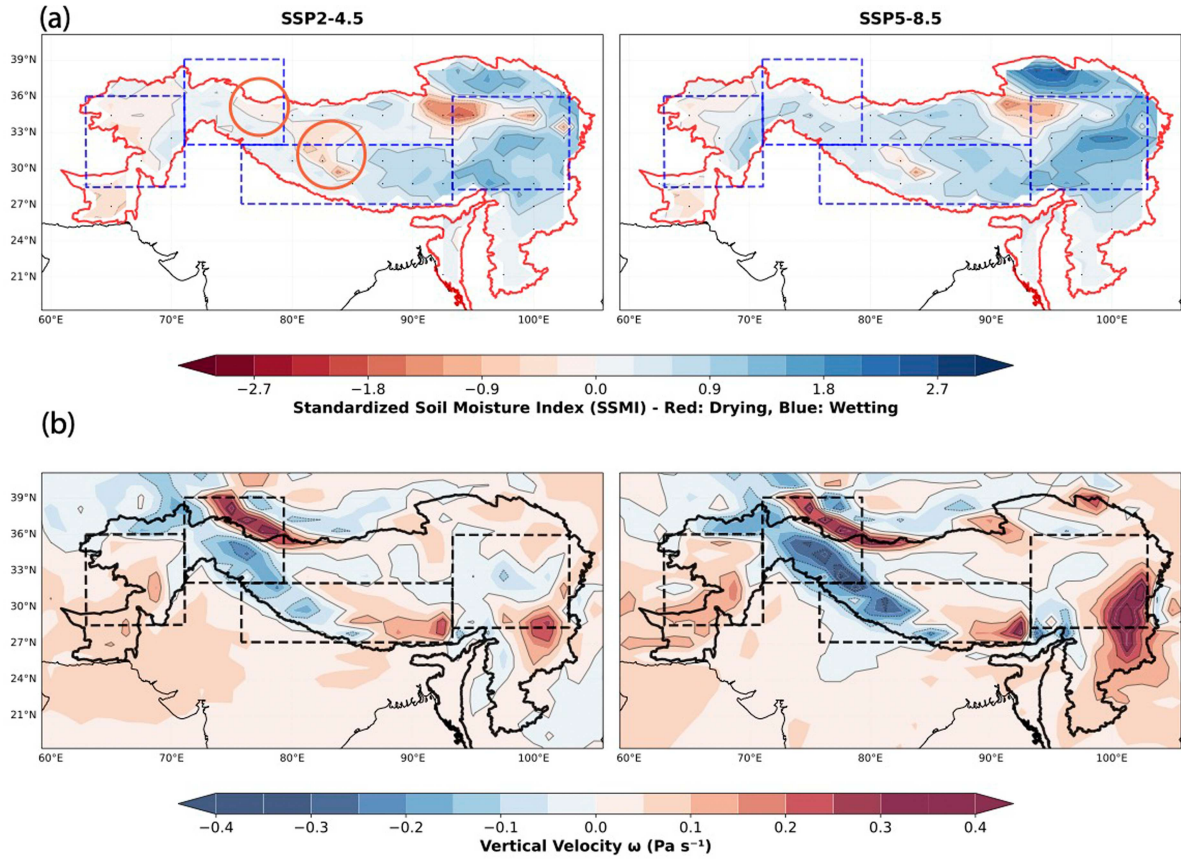


Figure 9. The figure shows (a) Standardized Soil Moisture Index (SSMI) for the upcoming 85 years (2015–2100). The dotted point represents the area that passes the 95% confidence level. (b) Changes in the vertical velocity (VV, defined as $-\omega$) at W500 hPa from the lowest to the highest emission SSPs from 2015 to 2100. Changes are estimated from the 1995–2014 mean of the CMIP6 multimodel ensemble. The red circles indicate the drying situation over Zone 1 and Zone 2.

SPI indicated normal to moderately wet conditions ($0 \leq \text{SPI} \leq 1.5$), whereas SSP5-8.5 showed moderately dry to moderately wet conditions ($-1.5 \leq \text{SPI} \leq 1.5$).

3.3. Soil moisture (SM) response to climate warming

Figure 9a shows the spatial changes in the SSMI under SSP2-4.5 and SSP5-8.5 at a 95% confidence interval. The moisture content is more oriented over the northeast part of the HKH region, and the SSI varies from 0.5 to 1.2. However, the Zone 1 shows modest values for even assuming negative values in some regions (varying from -2 to -1.6), which indicates dry conditions in the future. The SWI varies from a dry to wet pattern (8–24) in Zone 3, while it is a highly wet condition over Zone 4 (varies from 20 to 40), implying significant changes in the projected precipitation distribution but low agreement among the models under all emission scenarios (Figure S15). Moreover, the high value of the SWI over Zone 3 and Zone 4 may reflect the impact of large areas of frozen soil, snow cover, and melting processes. Meanwhile, SWI significantly varies from 0 to 12, representing high model agreement in Zone 1 and Zone 2, indicating the projected dry stage over the region under the highest emission scenarios.

3.4. Possible physical mechanisms for changing the hydrological extremes

We first examine the changes in the VV at 500 hPa, under different emissions (Figure 9b). We found that, in general, there is a high updraft, which increases deep convection and is likely to lead to more extreme events over the Zones 3 and 4 regions, with moderate values of VV (-1.5 to -0.5 Pa/s) in the medium

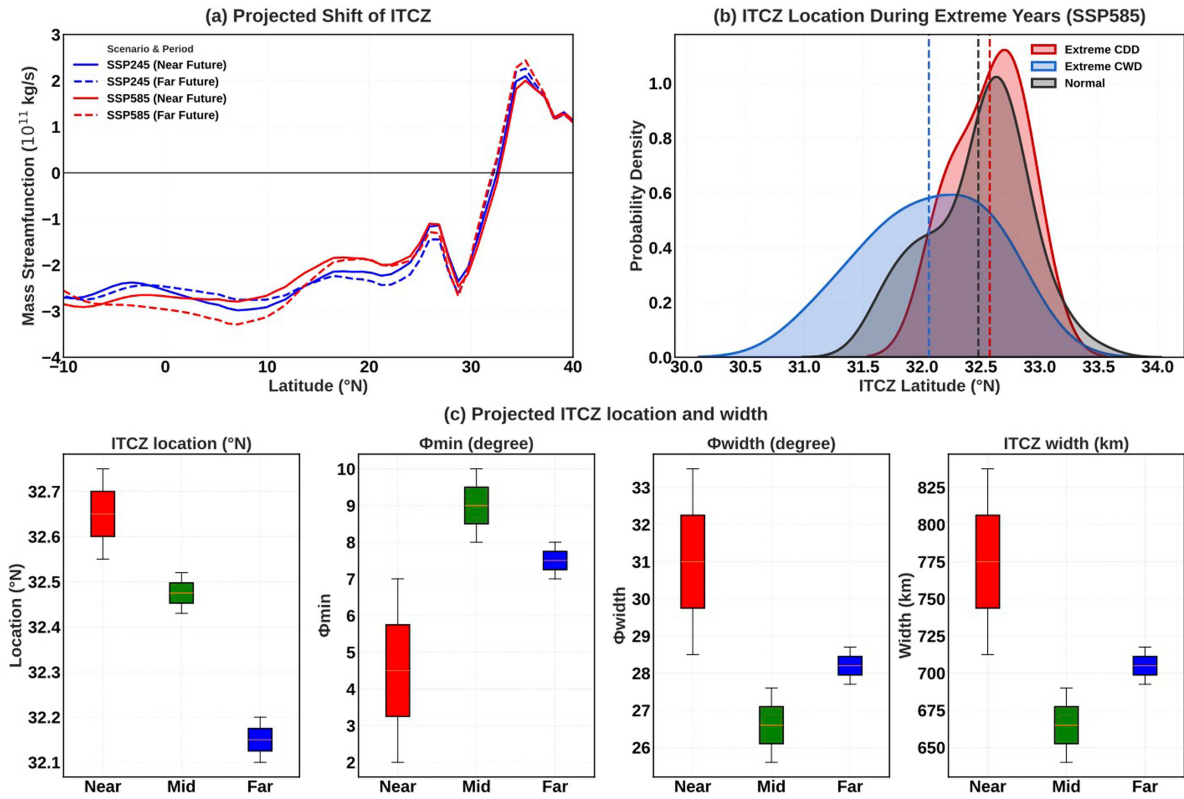


Figure 10. Projected change in the ITCZ for Zone 3. (a) The vertical average zonal mean meridional stream function from 700 to 300 hPa under SSP2-4.5 and SSP5-8.5. (b) ITCZ latitude during normal years and extreme years, indicating the interannual variability during extreme years. (c) The boxplot shows the projected ITCZ location and width for low to high emissions.

(SSP2-4.5) emission scenarios, while the highest emission scenario SSP5-8.5, exhibits moderate to high values (-2.5 to -0.5 Pa/s) of VV. However, Zone 1 shows less convection with estimated VV values ranging from 0 to 0.15. The western Himalayan Mountain range of Zone 3 shows strong moisture ($VV > -0.20$) convection in SSP5-8.5, which could be associated with more cloud formation and lead to more extreme situations over the region.

Next, we analyze the changes in the longitude of the ITCZ in different emission scenarios using a zonal mean meridional mass stream function (Figure 10). The study found a shift in the ITCZ position of approximately 0.2° latitude to the north for higher emission, SSP5-8.5, in comparison to the current medium emission SSP2-4.5 for the central Himalayan region Zone 3. However, the highest emission scenario (SSP5-8.5) intriguingly indicates a more southward shift of the ITCZ position in both mid- and far-future periods relative to the near-future period (Figure 10a). To be precise, the far-future (32.1°) changes in the ITCZ relative to the near-future (32.7°) reveal changes in the ITCZ position with a southward shift of approx. 0.6° latitude (approx. 72 km). Additionally, a similar signature occurs in SSP2-4.5, as the ITCZ position shows an approximately 0.3° latitude (approx. 38 km) shift to the south in the far future as compared with the near future (Figure 10a). When we examine the projected shift in the ITCZ with wet and dry years, we found that during wet years extremes ITCZ shifts southward from its normal year position, while during dry year extremes indicate, the ITCZ shifts more northward (Figure 10b).

Next, to quantify it further, we computed the ITCZ location and width of the ITCZ (Figure 10c) over the Zone 3 region for the medium emission scenario. Figure 10c shows a boxplot representation of the projected ITCZ location (degrees) and width (kilometres) under medium emission scenarios SSP2-4.5. In the near future, the ITCZ is projected to widen, spanning approximately 28.5° – 33.5° under medium emission scenarios; however, during the mid-century period, it may shift southward, ranging from 27.6° to 25.6° (Figure 10c). However, in the highest emission scenarios, SSP5-8.5, the difference between the

Table 3. ITCZ Location and Width from latitude 10°S–40°N.

SSPs scenarios	Temporal Scale	ITCZ location (°N) Φ_{ITCZ}	Φ_{max} (°N)	Φ_{min} (°N)	$\Phi_{max} - \Phi_{min} = \Phi$ (°N)	ITCZ width Φ_{width} (km)
SSP2-4.5	Near	32.5	35.5	7	28.5	712.5
SSP5-8.5		32.7	35.5	2	33.5	837.5
SSP2-4.5	Mid	32.4	35.6	8	27.6	690
SSP5-8.5		32.5	35.6	10	25.6	640
SSP2-4.5	Far	32.2	35.7	8	27.7	692.5
SSP5-8.5		32.1	35.7	7	28.7	717.5

maximum and minimum latitudes is 33.5° during the near future, indicating the wider conditions and leading to a drought (dry) situation over the region, while during the far future, it has been examined as a narrow (wet) approximate 28.7° as compared with the near future (Table 3). The narrow ITCZ (far from the future) leads to more extreme precipitation (wet condition), while the wider ITCZ (near the future) leads to drought (dry condition) over the region. These findings show complex interactions among the ITCZ, the Hadley cell, and vertical convection patterns, affecting precipitation over the Himalayan region under future warming scenarios.

4. Discussion

The present day precipitation extremes in the HKH region strongly depend on soil moisture dynamics, large-scale atmospheric circulation patterns (Attada et al. 2018), and the location and width of the ITCZ (Kad and Ha 2023), which collectively modulate precipitation intensity and variability. Understanding these observed drivers provides a mechanistic framework to better interpret the projected future changes. This study demonstrates that the areal mean precipitation over the HKH region increases consistently with increasing greenhouse gas emissions across all SSP scenarios. The projection for Zone 1 indicates a decline in annual precipitation, which is expected to impact millions across South Asian countries, posing socio-economic challenges for adaptation and mitigation (Fan et al. 2021). However, the increasing trend in precipitation over Zone 2 may be influenced by the moisture flux wind circulations related to wintertime precipitation in alliance with extended variations of synoptic western disturbances (Nischal et al. 2024).

The projected scenarios show extremes that are likely to intensify with global warming, indicating Clausius–Clapeyron theory, i.e., the amount of atmospheric moisture increases exponentially with rising temperature (Fowler et al. 2024). A pronounced shift from wet to dry conditions is projected over Zone 3, consistent with Zhang et al. (2024). These changes are likely driven by global warming, which weakens and delays monsoon circulation, a prominent factor behind the projected hydroclimate changes over the central Himalaya (Zhao and Dai 2022). This may be due to the future rising temperature contributing to glacier mass loss, which increases meltwater flow into downstream rivers (You et al. 2017). Such changes can influence regional hydrological cycles and, in turn, affect monsoon dynamics by altering land-atmosphere interactions, moisture availability, and surface energy balance (Immerzeel et al. 2010; Pritchard 2019). Meanwhile, Zones 1 and 2 are projected as drier stages in the future under the highest emission scenarios, and similar findings resulted in Zhai et al. (2020) for the northwest HKH regions, which are expected to experience longer and more frequent droughts. This finding is consistent with previous research concluding that global warming is likely to accelerate the evapotranspiration rate under SSP2-4.5 and SSP5-8.5, which leads to the depletion of soil moisture, which increases the frequency of soil moisture-based drought events in Central Asia (Zhao and Dai 2015; Hua et al. 2022).

The VV was found to be more convective in nature over Zone 2, while Zone 4 was more dominant during the summer monsoon season (JJAS). These changes in VV may be controlled by the thermodynamic amplification factor and stronger moist convection under future warming climates (Del Genio et al. 2007; Kim et al. 2022). Hari et al. (2020) observed a northward strengthening and propagation of the ITCZ since 2002 using reanalysis datasets. The physical link between the ITCZ width and HKH precipitation operates through modulation of the Hadley circulation strength. A wider ITCZ reflects weakened meridional overturning and reduced poleward moisture transport, resulting in drier conditions over mid-latitude mountainous regions. Conversely, ITCZ narrowing intensifies the meridional circulation, enhancing moisture convergence over the HKH (Byrne and Schneider 2016; Kad and Ha 2023).

Under higher emission scenarios, the ITCZ shifts more substantially, likely driven by the moist component of vertical velocity, which is governed by adiabatic heating from deep convection (Nie et al. 2018). The subsequent ITCZ narrowing during the mid-to-far future re-intensifies meridional overturning and moisture flux, driving the projected shift to wetter conditions. Physically, a wider ITCZ corresponds to a weakened Hadley cell that reduces poleward moisture transport to the mid-latitudes, suppressing precipitation over HKH Zone 3 during the near future. The subsequent ITCZ narrowing during the mid-to-far future re-intensifies meridional overturning and moisture flux, driving the projected shift to wetter conditions. While this large-scale framework is consistent with Byrne and Schneider (2016) and Kad and Ha (2023), regional monsoon dynamics, including South Asian monsoon variability and westerly jet interactions, are complementary drivers that warrant further investigation in future work.

4.1. Limitations and uncertainties

Despite the strengths of the CMIP6 multi-model ensemble and the use of SSP scenarios, several limitations and uncertainties remain in this study. First, the CMIP6 models exhibit systematic biases over the HKH region. A cold bias of approximately 2–4 °C is evident in Zones 1 and 2, and summer precipitation is overestimated by 10%–25% in Zone 4, as reflected in the Taylor Skill Score analysis (Table S1). Although quantile mapping bias correction reduces these biases for the historical period, the assumption of stationarity in the correction factors introduces residual uncertainty for future projections. Second, the re-gridding of all model outputs to $1^\circ \times 1^\circ$ coarsens the representation of steep elevation gradients that are critical for orographic precipitation enhancement over the HKH. Convective processes and localised precipitation extremes are therefore likely underestimated in the ensemble mean fields. The intermodel spread in annual precipitation increases systematically from west to east across the HKH domain, with the ensemble standard deviation rising from 79 mm yr⁻¹ in Zone 1 to 307 mm yr⁻¹ in Zone 4 (Supplementary Figures S1–S3). The Zone 1 exhibited the highest relative uncertainty (32%), reflecting fundamental model disagreement in simulating arid, high-elevation climates. Importantly, Zone 1 is the only region where models systematically underestimate precipitation, which is consistent with the documented inability of coarse resolution GCMs to capture orographic enhancement by wintertime western disturbances (Palazzi et al. 2013; Cannon et al. 2015). Third, the projected shifts in the ITCZ position and width are not fully consistent across all emission scenarios. This inconsistency likely reflects differences in model sensitivity to aerosol forcing and internal climate variability (Byrne et al. 2018; Jiang et al. 2023). Sub-degree shifts in large-scale circulation features are particularly susceptible to uncertainty in coarse-resolution models (e.g., Adam et al. 2016; Byrne et al. 2018). Fourth, the soil moisture analysis is limited to the 10 cm surface layer, which captures short-term drought signals but may not fully represent deeper soil moisture storage relevant to perennial vegetation and groundwater recharge. Finally, high-resolution model simulations such as those from the High Resolution Model Intercomparison Project (HighResMIP; Haarsma et al. 2016) could help reduce uncertainty in orographic precipitation and circulation dynamics and are recommended for future work over this complex terrain.

5. Summary and conclusion

The findings from our research are crucial in deepening our understanding of the projected changes in hydroclimate extreme events with coherent soil moisture variations in low- to highest-emission scenarios over the HKH. The primary conclusions from this analysis are outlined as follows:

- 1) The drought mitigation and water management strategies are required for Zones 1 and 3, where declining precipitation and increasing dry spell frequency under the current emission trajectory SSP2-4.5 threaten agricultural productivity and water security across millions of people in the western and central HKH.
- 2) The flood resilience infrastructure planning must be emphasized for Zones 2 and 4, where extreme precipitation under higher emission SSP5-8.5 will intensify significantly in the far future, leading to an increasing risk to downstream city populations and infrastructure.

- 3) The projected shifts in the ITCZ position and width reveal that large-scale meridional circulation changes are a primary driver of HKH hydroclimate variability. A wider ITCZ in the near future decreases the precipitation over Zone 3, weakening poleward moisture transport, which may lead to dry conditions, while subsequent narrowing in the far future intensifies moisture convergence and drives wet conditions.

This study provides valuable insights into the future projection of HKH hydroclimate extreme events. It enhances our understanding of the intricate topography and associated extremes due to deep convection under different emission scenarios. Therefore, it is important to implement an advanced early warning system in the HKH region to aid policymakers and stakeholders in mitigating these climate risks.

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Author contributions

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Data availability statement

The gauge-based precipitation and CMIP6 datasets are freely available at <http://aphrodite.st.hirosaki-u.ac.jp> and the Earth System Grid Federation (<https://esgf-node.llnl.gov/search/cmip6/>), respectively. Land use and land cover data were obtained from publicly available sources: ESRI Global Land Cover (<https://livingatlas.arcgis.com/landcover>) and NOAA CDR Land Cover from AVHRR imagery (<https://www.ncei.noaa.gov/products/land-cover>). The NCL, Python, and R scripts used for data analysis in this study are available from the corresponding author upon reasonable request.

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